CSE142 Object Oriented Programming

Sample idea for semester projects.

The following list are just sample ideas for your CSE142 course project. You can come up with your idea. You must implement object-oriented principles in your semester project.

The project is to be done in a group so that you can learn to work in collaboration. Only three students at maximum are allowed in a group.

1. **Library Management System:**
    - Create classes for books, users, and transactions.
    - Implement features for adding, updating, and deleting books.
    - Manage user accounts, borrowing, and returning books.
2. **Student Information System:**
    - Design classes for students, courses, and grades.
    - Implement functionalities like adding students, assigning courses, and recording grades.
3. **Bank Management System:**
    - Create classes for customers, accounts, and transactions.
    - Implement features such as account creation, deposit, withdrawal, and transaction history.
4. **Online Shopping System:**
    - Design classes for products, shopping carts, and users.
    - Implement functionalities like adding products to a cart, managing user accounts, and processing orders.
5. **Inventory Management System:**
    - Create classes for products, suppliers, and inventory.
    - Implement features for tracking product quantities, managing suppliers, and generating reports.
6. **Car Rental System:**
    - Design classes for cars, customers, and rental transactions.

- Implement functionalities like renting and returning cars, managing customer accounts, and tracking available cars.

7. **Employee Payroll System:**
    - Create classes for employees, departments, and payroll.
    - Implement features such as calculating salaries, managing employee details, and generating payroll reports.
8. **Hospital Management System:**
    - Design classes for patients, doctors, and medical records.
    - Implement functionalities like scheduling appointments, managing patient information, and tracking medical history.
9. **Hotel Reservation System:**
    - Create classes for rooms, guests, and reservations.
    - Implement features for booking rooms, managing guest details, and processing reservations.
10. **Social Media Network:**
    - Design classes for users, posts, and comments.
    - Implement functionalities like creating posts, commenting on posts, and managing user profiles.

11. **Chess with AI:**
    - Design classes for the chessboard, pieces, players, and the game itself.
    - Implement the rules of chess, including movement validation and check/checkmate conditions.
    - Develop an AI opponent with varying levels of difficulty.
12. **Cryptocurrency Exchange:**
    - Create classes for different cryptocurrencies, user accounts, and transactions.
    - Implement functionalities for buying, selling, and exchanging cryptocurrencies.
    - Include a secure authentication system for user accounts.
13. **Disease Predictor:**
    - Design classes for patients, symptoms, and diseases.

- Implement an algorithm to predict potential diseases based on symptoms provided by the user.
- Provide information on preventive measures and treatment options.

14. **Path Finding Visualizer:**
    - Develop classes for a grid, nodes, and algorithms (e.g., Dijkstra's, A*).
    - Create a visual representation of pathfinding algorithms on a grid.
    - Allow users to draw obstacles, start, and end points for pathfinding visualization.
15. **Supermarket Management System:**
    - Design classes for products, customers, and transactions.
    - Implement features for managing inventory, processing sales, and generating sales reports.
    - Include a user-friendly interface for employees.
16. **E Bank Management System:**
    - Create classes for accounts, transactions, and customers.
    - Implement functionalities like account creation, balance inquiries, and fund transfers.
    - Ensure security features, such as password protection and encryption.
17. **Management System for Construction and Building Companies:**
    - Design classes for projects, employees, and tasks.
    - Implement features for project management, employee assignment, and progress tracking.
    - Include functionalities for cost estimation and resource allocation.
18. **Console-Based Operating System:**
    - Develop classes for processes, memory management, and file systems.
    - Implement basic operating system functionalities in a console environment.
    - Include features like process scheduling and file operations.
19. **Cricket Score Management System:**
    - Design classes for teams, players, matches, and scores.
    - Implement features for scoring, player statistics, and match history.
    - Include options for live score updates during a match.

20. **Console-Based Game:**

- Create classes for game entities, levels, and players.
- Implement a simple console-based game, such as a text-based adventure or a basic strategy game.
- Include features like player interaction and scoring.